

Progettazione di Applicazioni Mobili

Mobile design patterns

Navigation

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Interaction design patterns

- Interaction design patterns are a way to describe solutions to common user experience problems in a specific context
- They document interaction models that make it easier for users to understand an interface and accomplish their tasks
- Distinct design conventions for the main mobile OSs have become more formalized over time

- Benefits of using interaction design patterns include:
 - Teaching some best practices and common approaches, capturing collective wisdom of designers across many uses and scenarios
 - Giving teams a common language, reducing misunderstandings that arise from different vocabulary
 - Eliminating wasted time spent "reinventing the wheel", reducing time and costs in the design and development lifecycle
 - Ensuring users have a consistent and predictable experience within an application or service

Navigation patterns

- Navigation patterns are among the most important in the design of mobile applications
- Following common design patterns for navigation will ensure that people can find and use the valuable features in your application
- Good navigation, like good design, is invisible. Applications with good navigation just feel simple and make it easy to accomplish any task

- Mobile navigation patterns can be used to support both ***primary*** navigation and ***secondary*** navigation
 - Primary navigation patterns are those related to accessing the main features of an application (e.g., navigating from one primary category to another, as with the top-level menus of a desktop application)
 - Secondary navigation patterns are those related to moving about within a selected module/section of an application
 - Most primary navigation patterns can often be used as secondary navigation patterns as well

Primary navigation

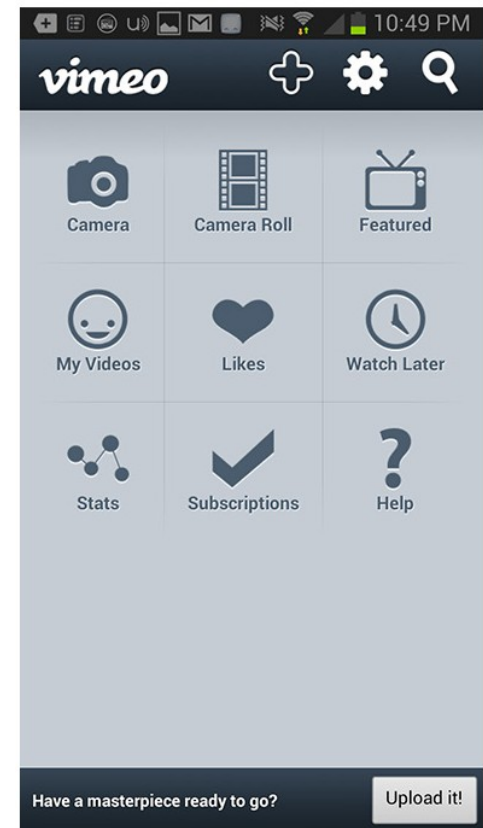
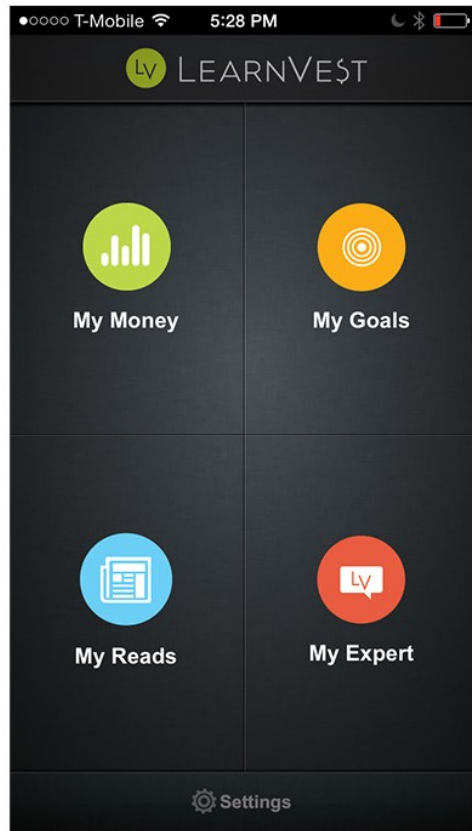
- Primary navigation includes two distinct types of pattern:
persistent and ***transient***
 - Persistent navigation patterns include those designs in which navigation options are immediately visible as soon as the user opens an app (and typically stay visible during interaction)
 - Transient navigation patterns include those designs in which navigation options must be explicitly revealed with some user action (e.g., a tap or gesture)
 - Transient navigation patterns are hidden until the user reveals them; they disappear when the user makes a selection

Persistent vs transient navigation

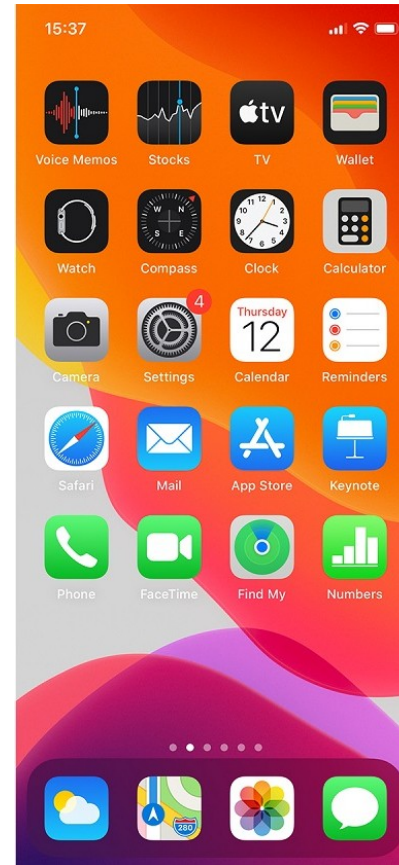
- If you answer “yes” to one or more of these questions, it’s probably best to use persistent navigation
 - Is your application “flat”? Are the menu categories equivalent in hierarchy, and are there just a few primary categories (i.e., three to five) in the app?
 - Do your users need the menu to be always visible for quick access?
 - Do the menu categories have status indicators, like the number of unread emails, for instance?

Persistent navigation: springboard

- The ***springboard*** design is a grid-based layout with options that act as launch points into the application

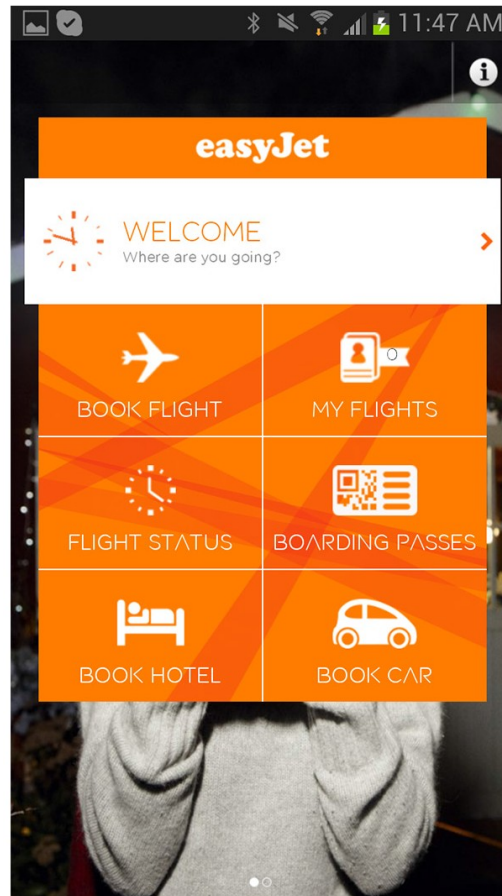
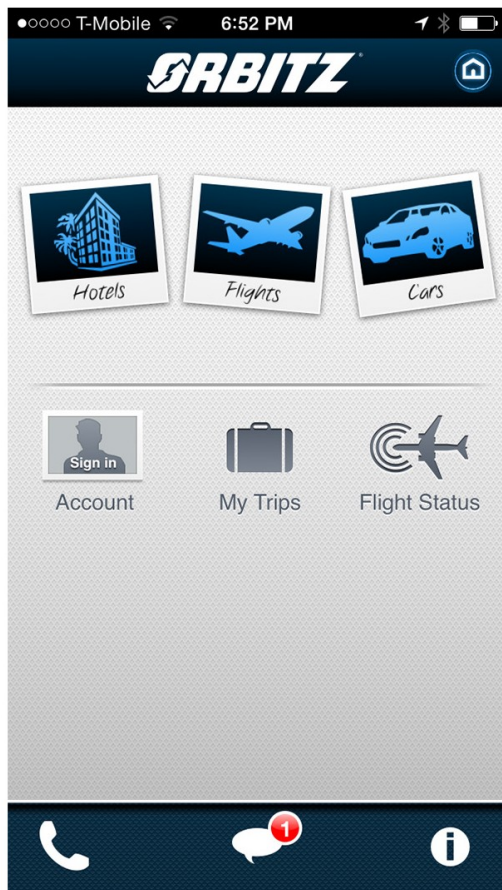


- Android and iOS use the springboard navigation pattern at the OS level (with paging or scrolling)



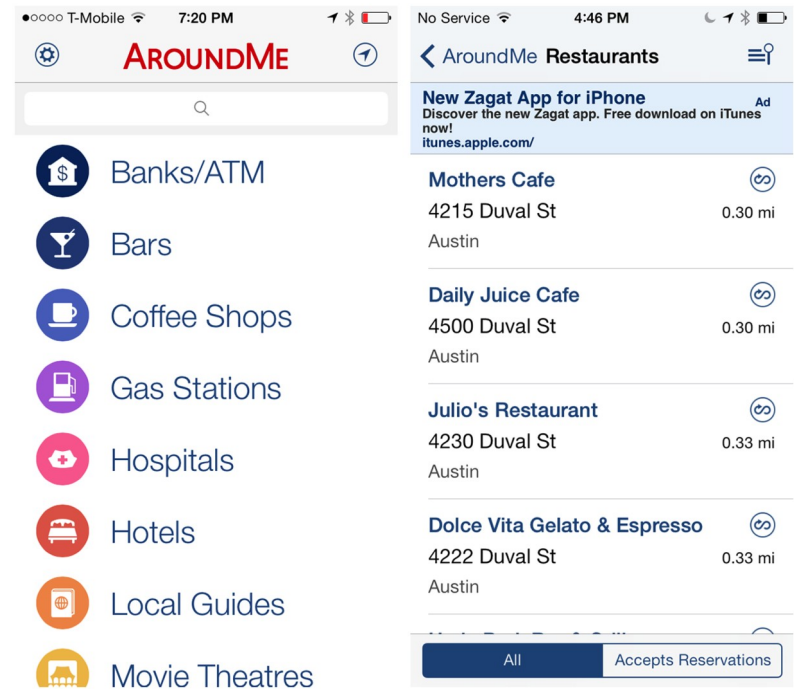
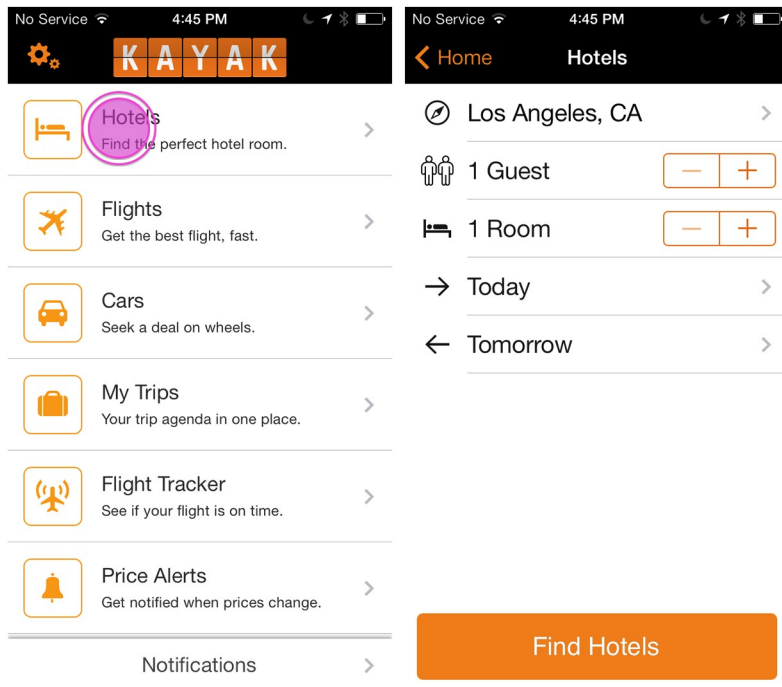
- A common springboard design displays up to 9 options in a 3x3 grid
 - By adding a page indicator (usually, through dots at the bottom), designers can provide additional options
- Today, the springboard pattern is not popular
- Drawbacks of the springboard pattern:
 - It flattens all options to the same level of importance
 - It uses the home screen space just for navigation

- By varying icon treatment and grid layout, it is possible to introduce visual hierarchy in the springboard design



Persistent navigation: list menu

- The ***list menu*** design displays navigation options as a vertical list



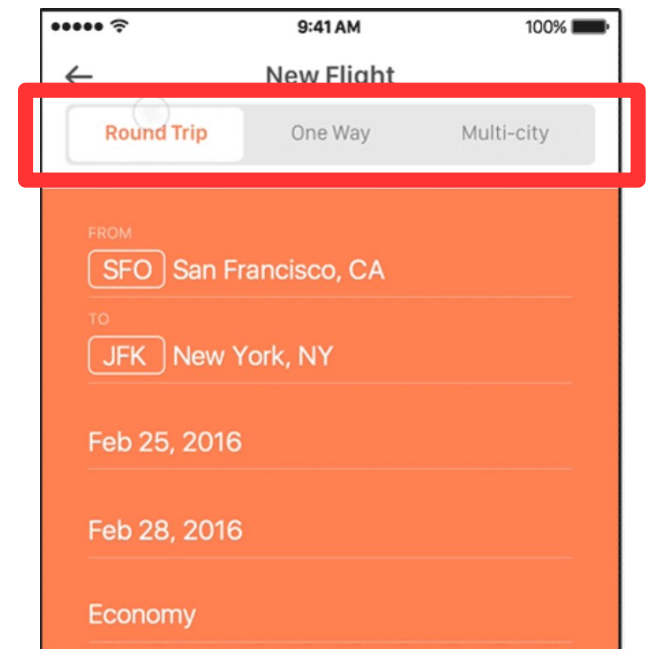
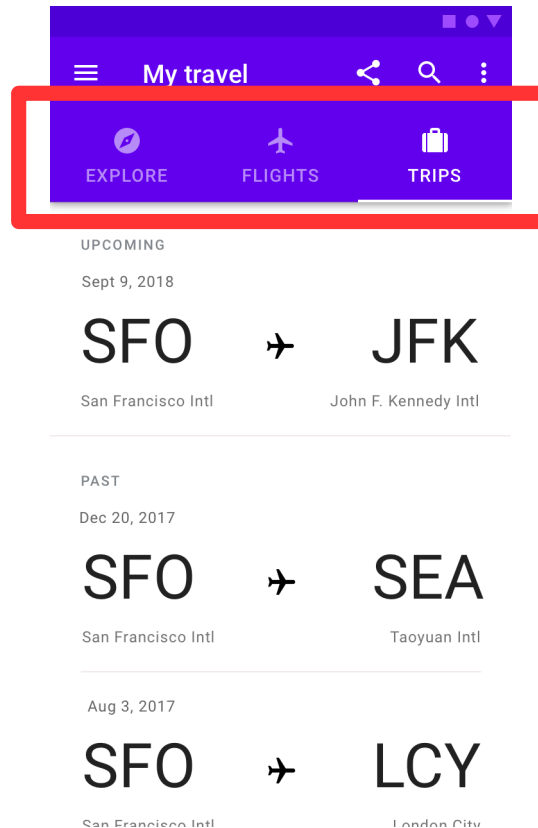
- List menus are typically used for navigating a hierarchical structure (in such case, there should be an option for the user to navigate back)
- They also work well for menus with long item names, and where items need descriptions as well as titles
- Today, the basic list menu pattern is not popular to support primary navigation
- Same drawbacks as the springboard pattern

Persistent navigation: tab menu

- **Tab menus** display navigation options as an horizontal bar at the top of the screen



Fixed tabs (Android)

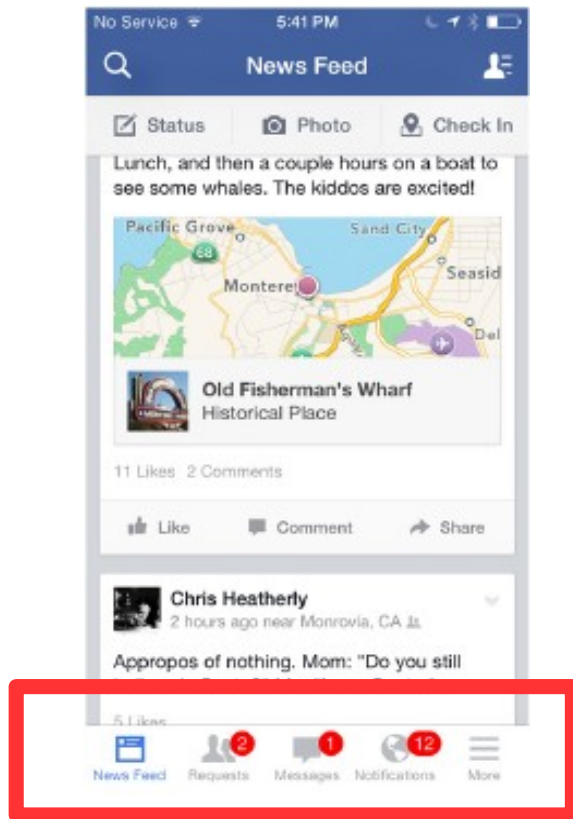


Segmented control
(iOS)

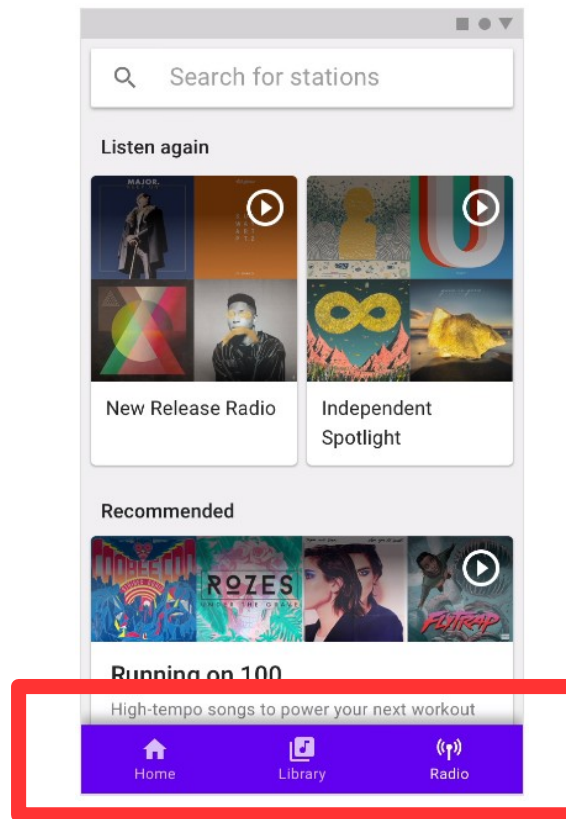
- Tab menus are always visible on the screen, and can't be moved out of the way
- The active tab should be highlighted
- Tab menus should allow the user to navigate between the views by swiping left or right on the content area
- You can use tab menus if:
 - You expect your users to switch views frequently
 - You have a limited number of up to three/four top-level views
 - You want the user to be highly aware of the alternate views

Persistent navigation: bottom bar

- **Bottom bars** display navigation options as an horizontal bar at the bottom of the screen



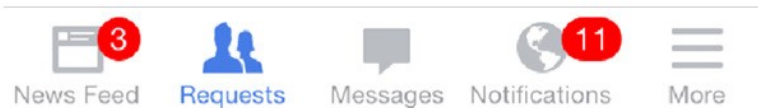
Tab Bar (iOS)



Bottom navigation
bar (Android)

- Bottom bars are recommended for navigating applications with a flat information structure
 - Ideally, it should remain visible everywhere (but it is common to hide it in secondary screens)
- If there is no space in the bottom bar for all navigation items, the bottom bar can slide or a More option can be provided as the last tab to access additional items
- Don't remove or disable an item in the bottom bar when its function is unavailable
 - If items are available in some cases but not in others, your interface becomes unstable and unpredictable

- Bottom bars and Toolbars look similar but they are functionally different
 - Bottom bars are used to navigate the main sections of the application
 - Toolbars presents the tools, or possible actions, for a specific screen

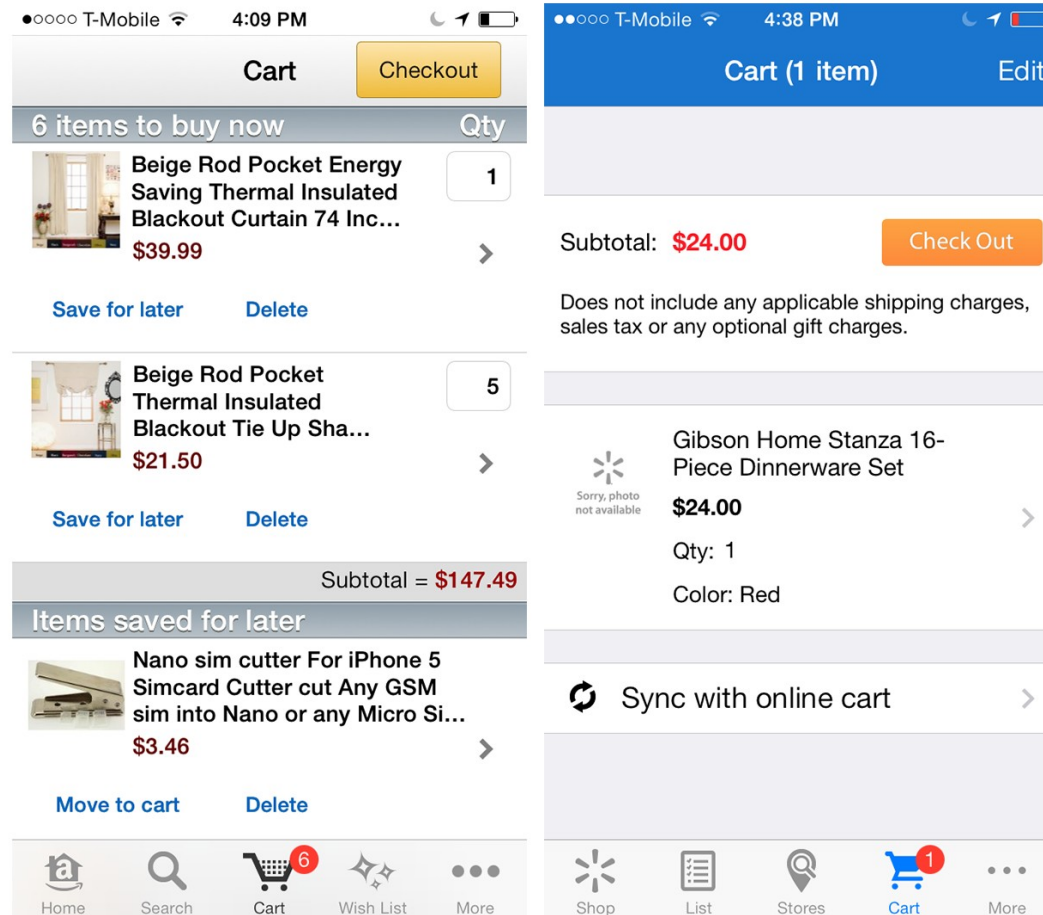


Bottom bar

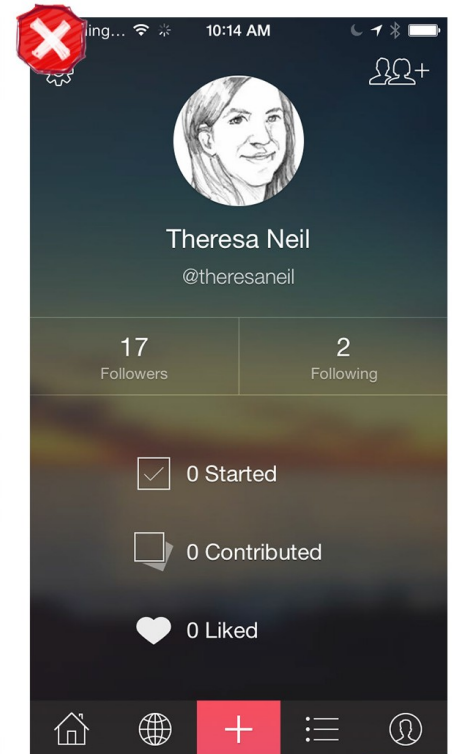
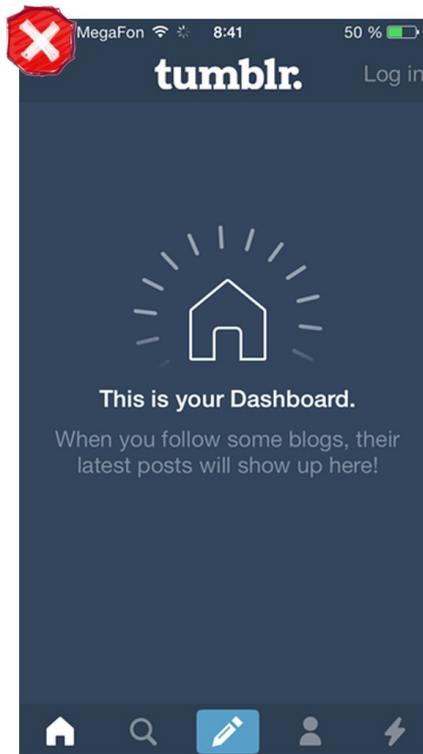
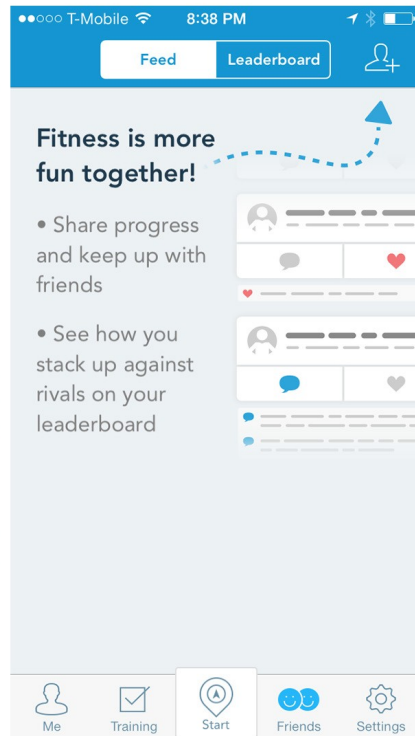


Toolbar

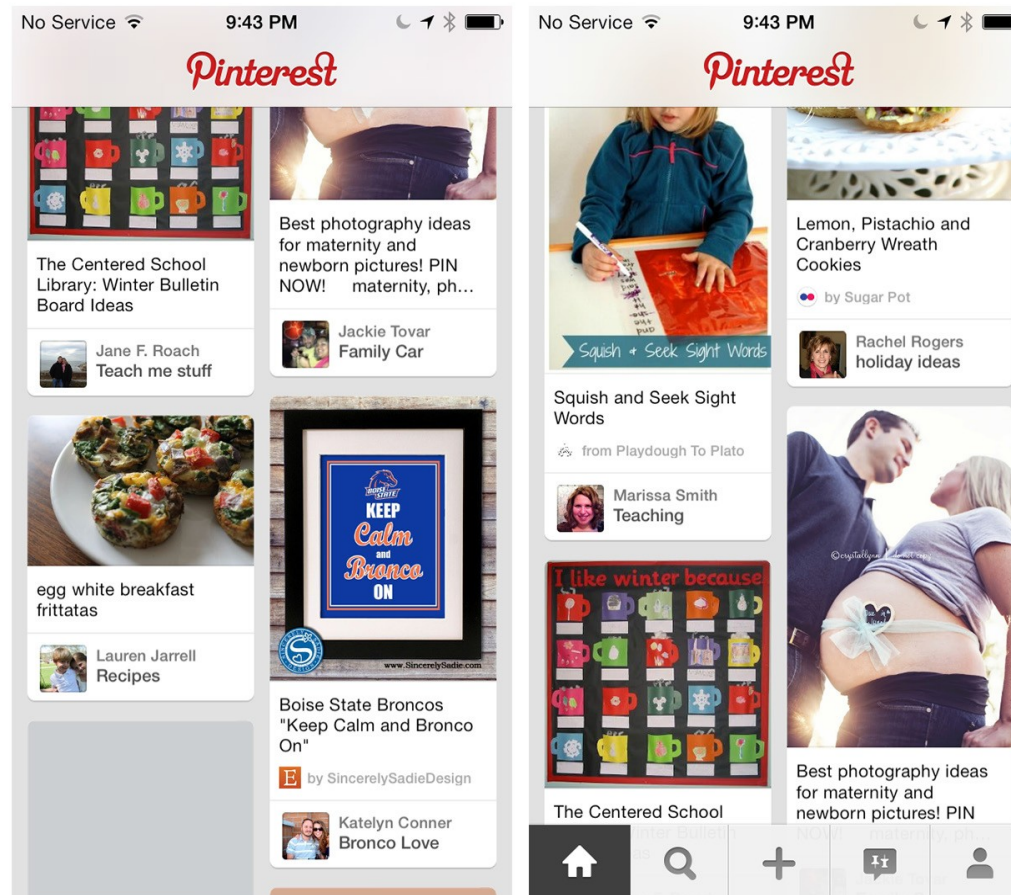
- The active item in the bottom bar should be highlighted



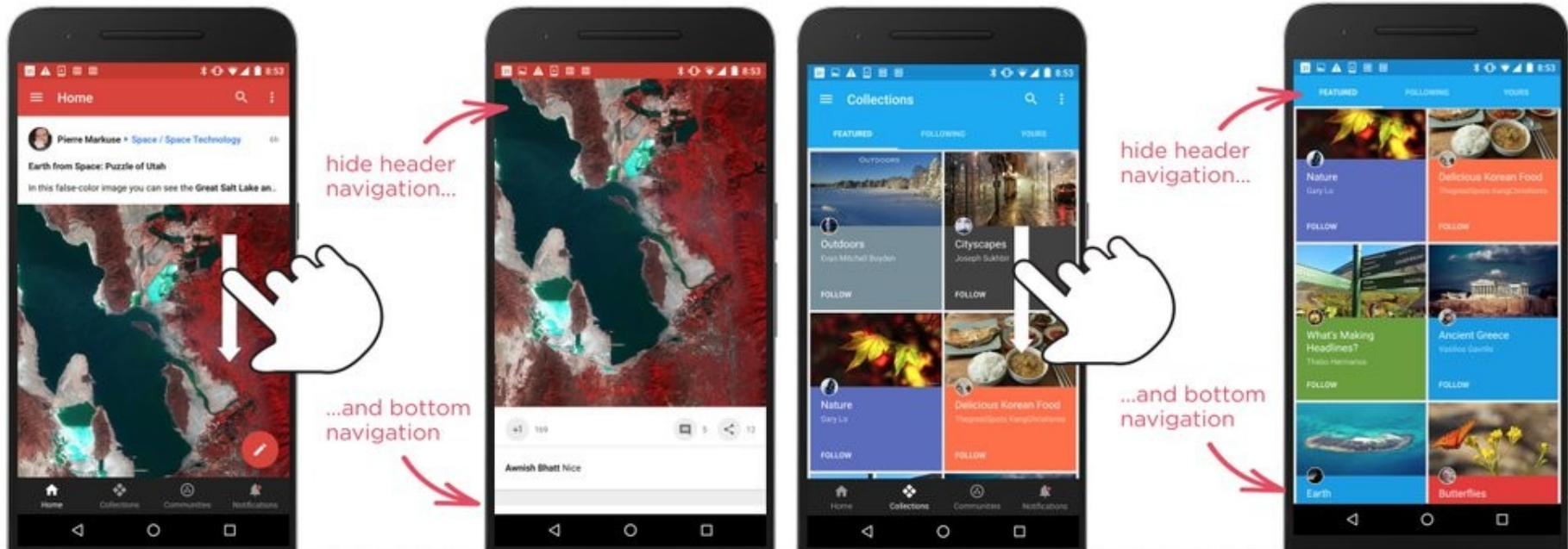
- Bottom bars can include a Call to Action pattern to make one item more prominent than the rest
 - make sure that selected items are not overshadowed by the visual emphasis on the Call to Action item



- The Retracting Bar design hides or collapses the bottom bar when the user is scrolling or swiping down through content

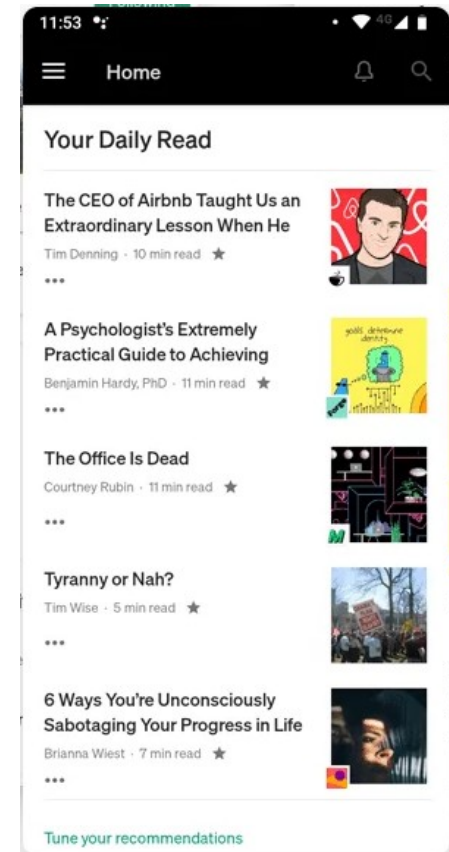
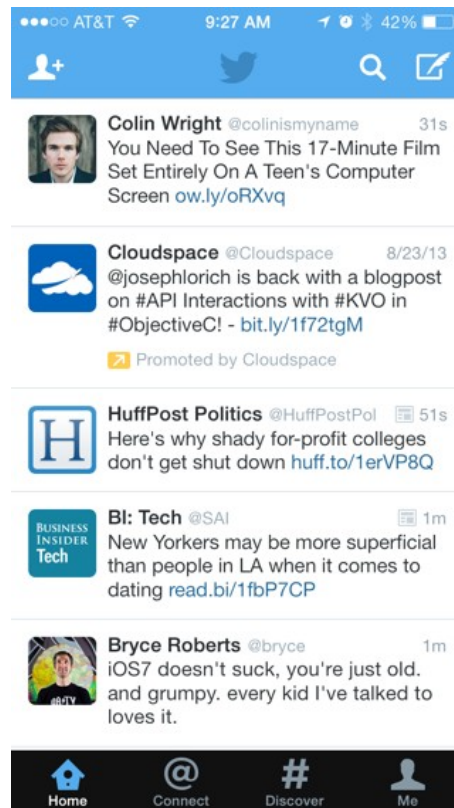
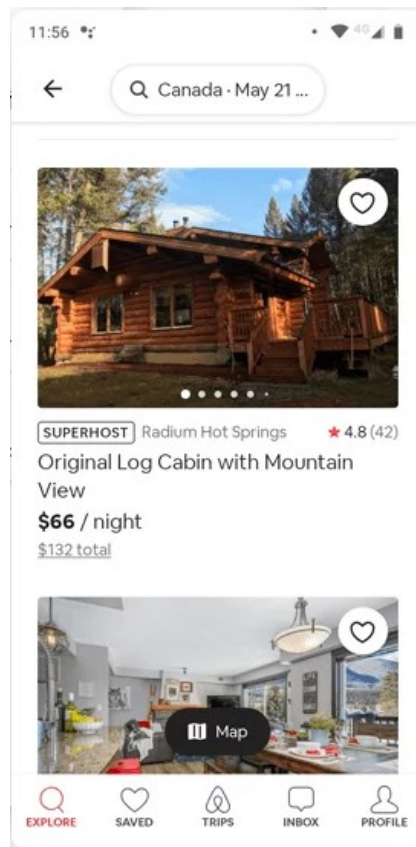


WHEN YOU SCROLL....



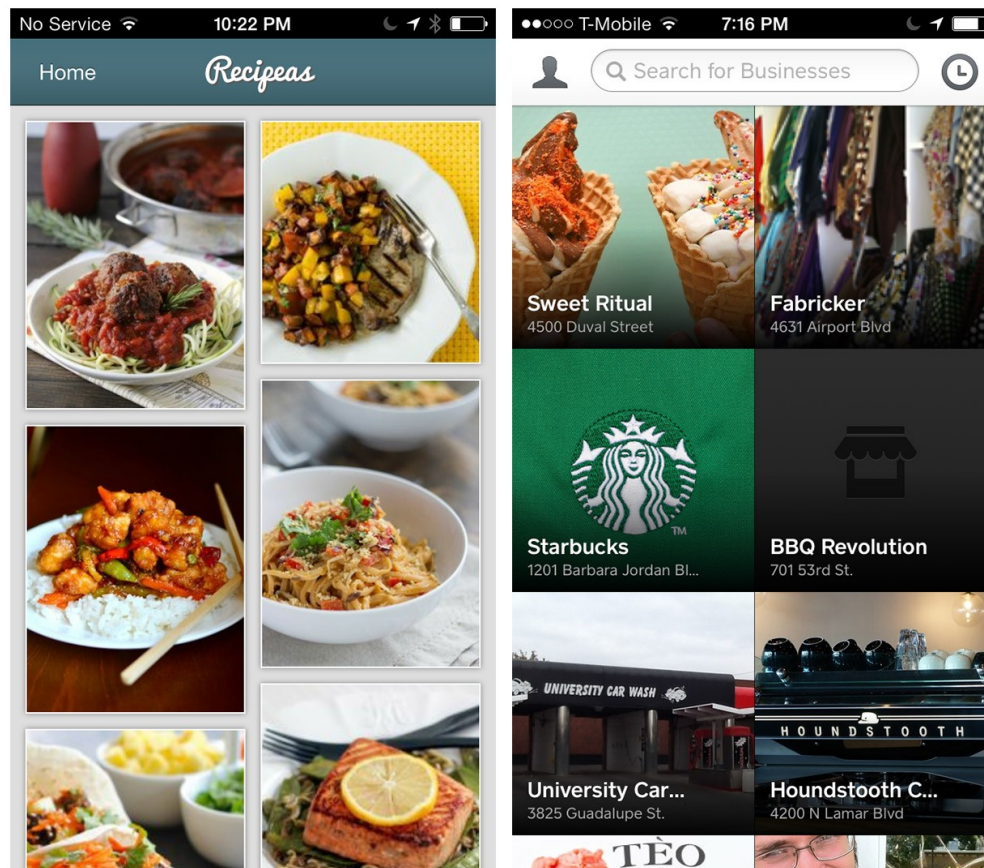
Persistent navigation: feeds

- The **feed** or **activity stream** pattern shows live content created by aggregating news, comments, posts, images, etc. in a (rich) list

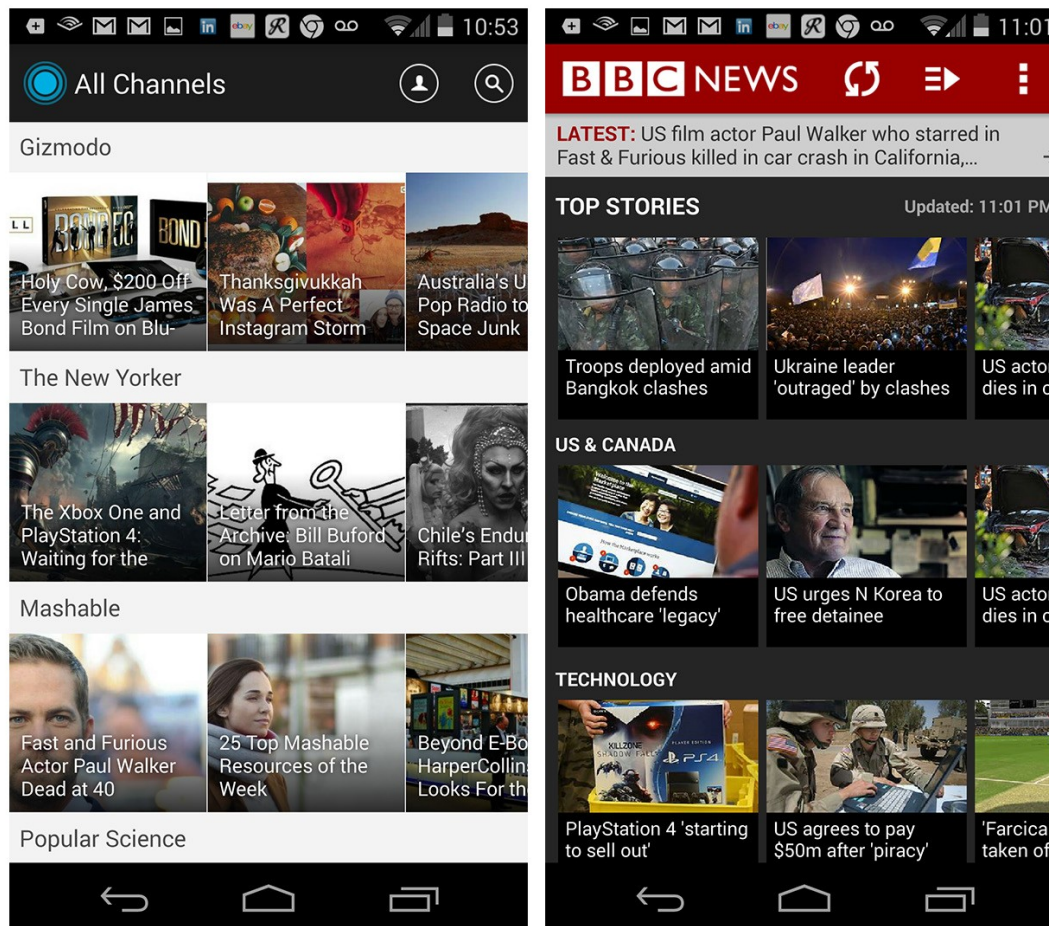


Persistent navigation: gallery

- The **gallery** pattern displays graphic pointers to live content like news stories, recipes, or photos, arranged in a (sliding) grid, a carousel, or a slideshow

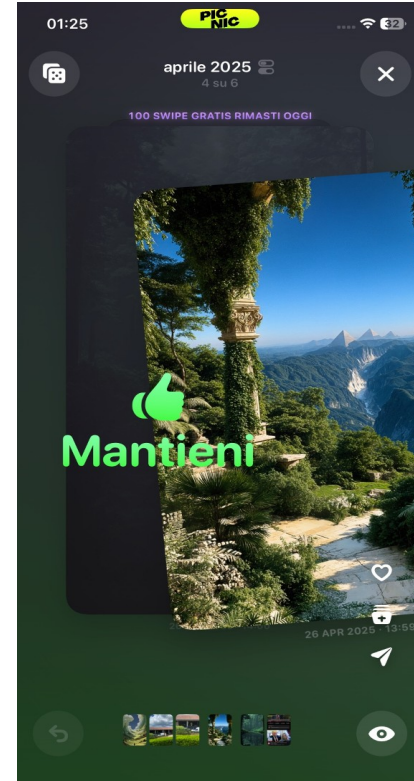
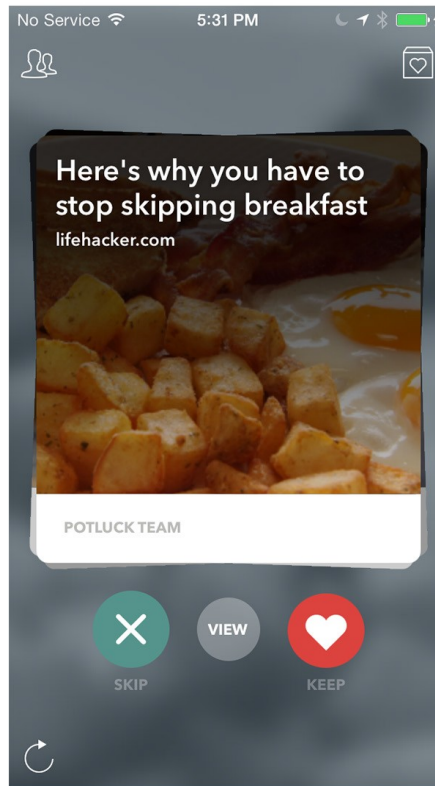
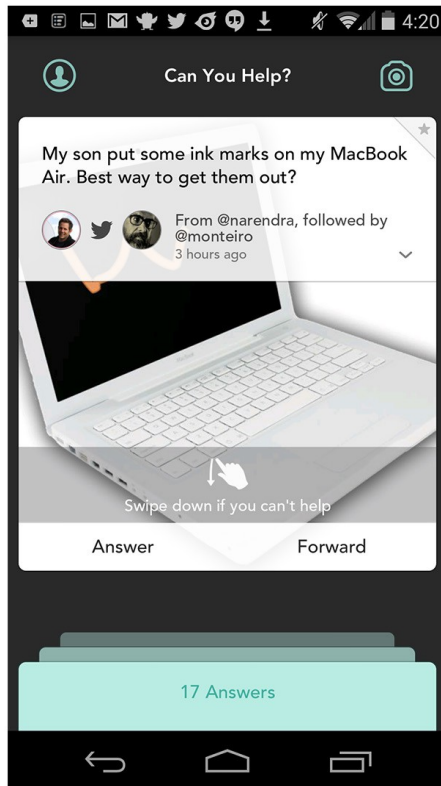


- The gallery pattern works best for showing frequently updated, highly visual content with no hierarchy



Persistent navigation: cards

- **Card** navigation is based on navigating content using a card deck metaphor, including common manipulations such as stacking, shuffling, discarding, and flipping



Persistent navigation: dashboard

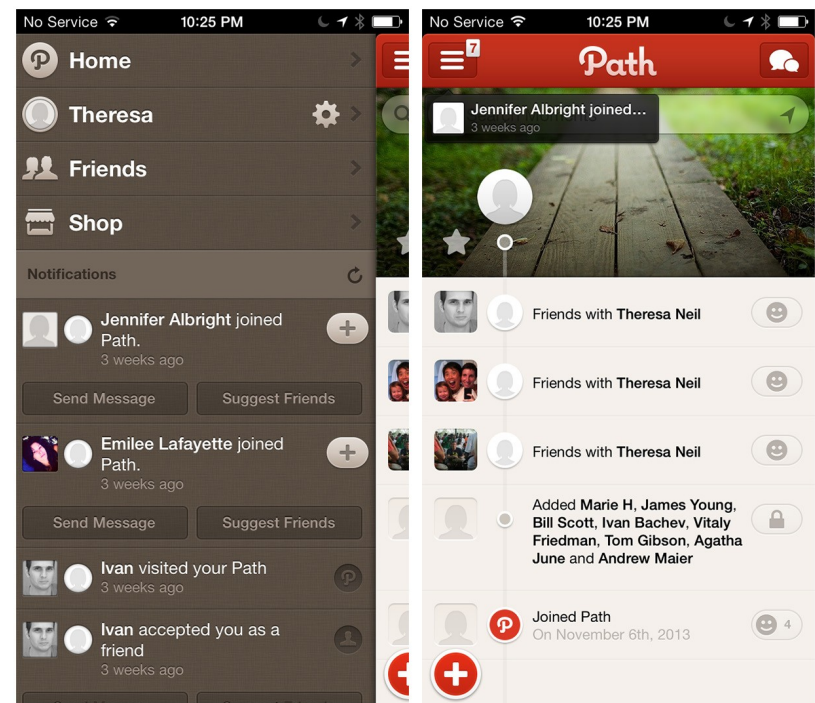
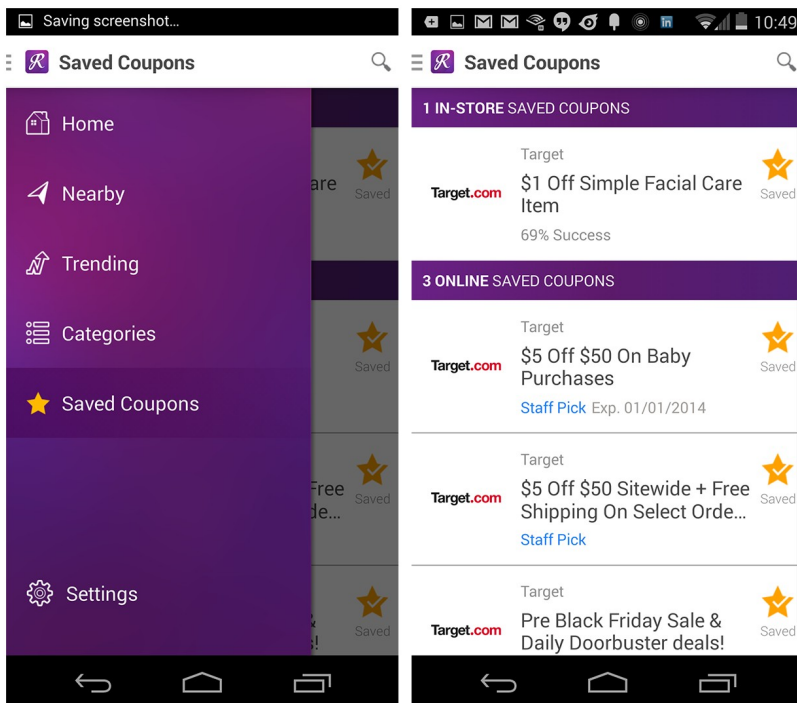
- A **dashboard** shows a summary of metrics/data that can lead to secondary screens with additional details



- The rules for providing navigation back to the dashboard are the same as with the list menu and springboard
- Use a dashboard only when it makes sense to use key metrics or data as launch points into the app
- Don't overload the dashboard; conduct research to determine which key metrics or data to include

Transient navigation: side drawer

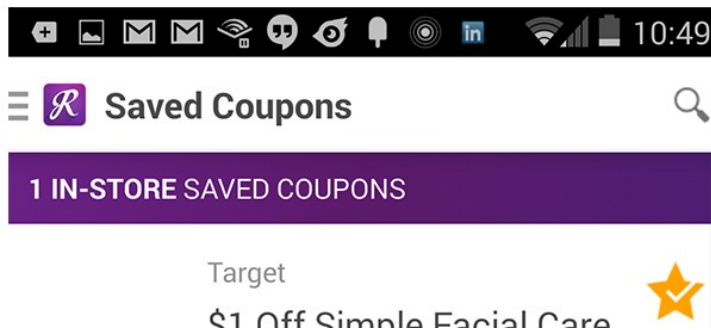
- There are two styles of ***side drawers***
 - *overlay* → the side drawer partially covers or overlaps the original screen content
 - *inlay* → the side drawer pushes the original screen content partially off-canvas



- The most popular position for the side drawer is on the left, but it can be on the right or both on the right and the left



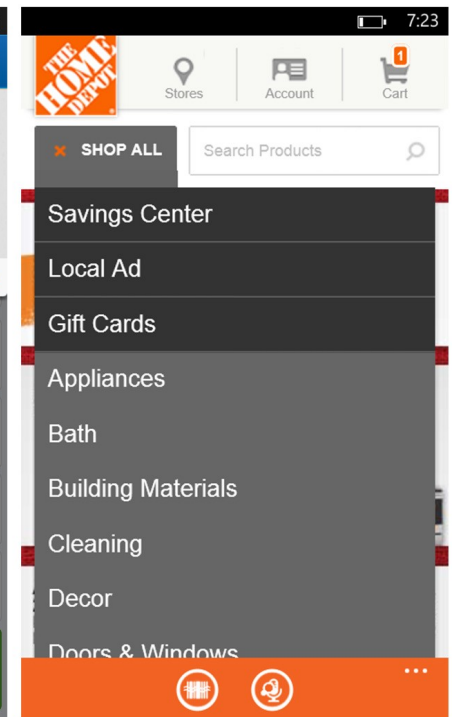
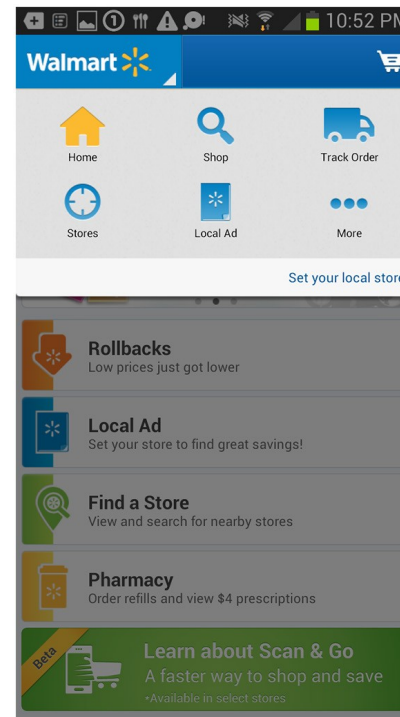
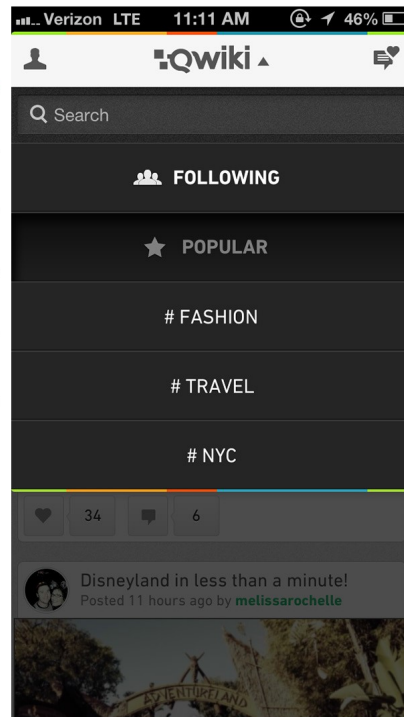
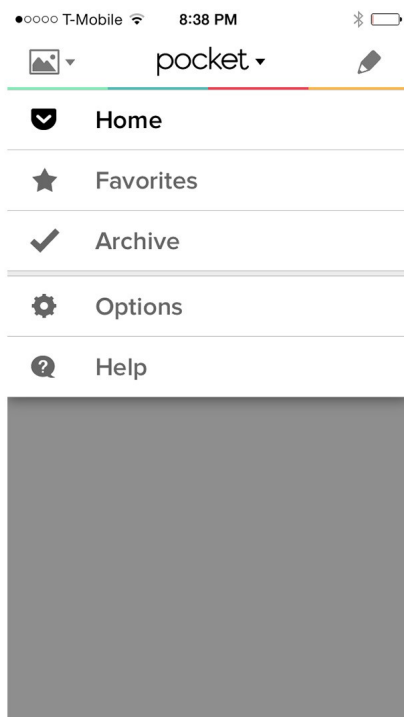
- Side drawer are not limited to navigation options, can be more than one level deep, but should not be overloaded with too many features
- The “hamburger” icon on the application bar signals availability of a navigation drawer



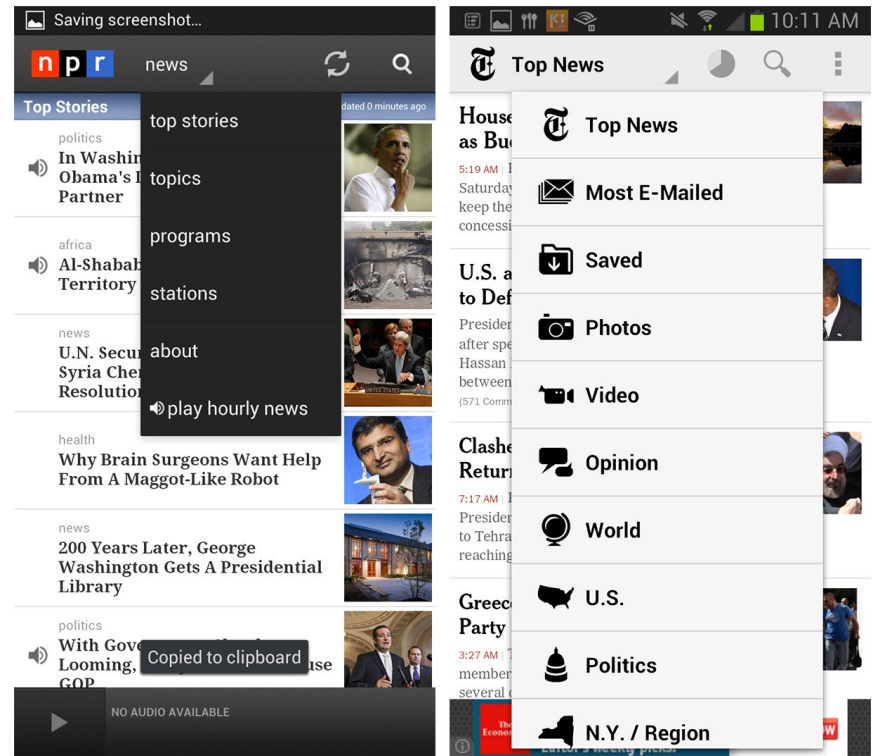
- The Android design guidelines recommend having the drawer open on first use so that users can see the menu and learn how to close the drawer

Transient navigation: toggle menu

- The ***toggle menu*** can be an inlay that pushes the content down below the menu or an overlay that appears as a layer above the content

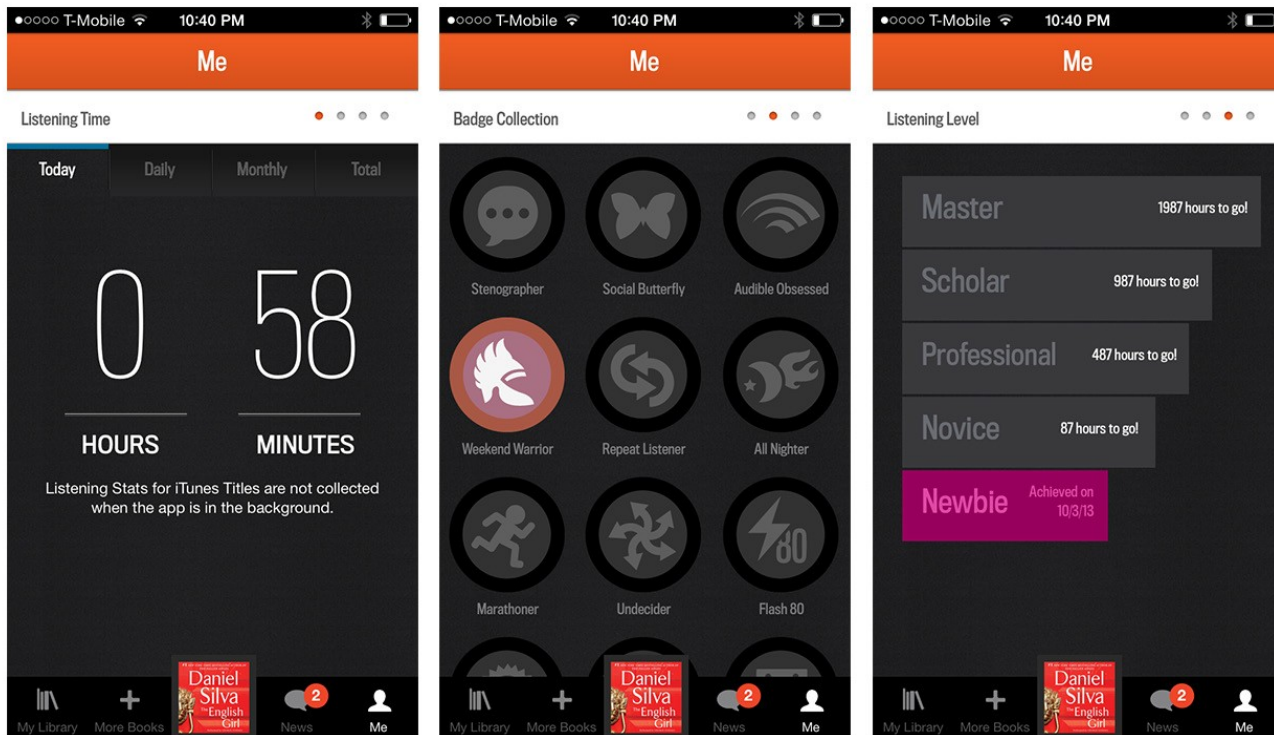


- Whatever gesture reveals the menu (tapping an icon, swiping, or panning) should also hide it
 - Tapping anywhere in the background should also hide the menu
- The menu shouldn't cover the whole screen, it should let the background peek through

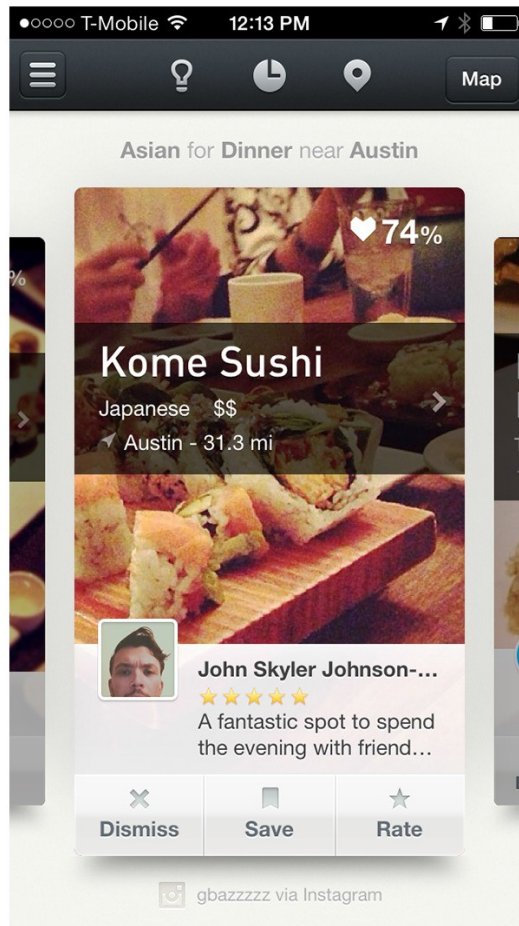


Secondary navigation: page swiping

- The ***page swiping*** pattern can be used to navigate quickly through content using the swipe gesture
- The most common way to communicate this navigation pattern is via page indicators

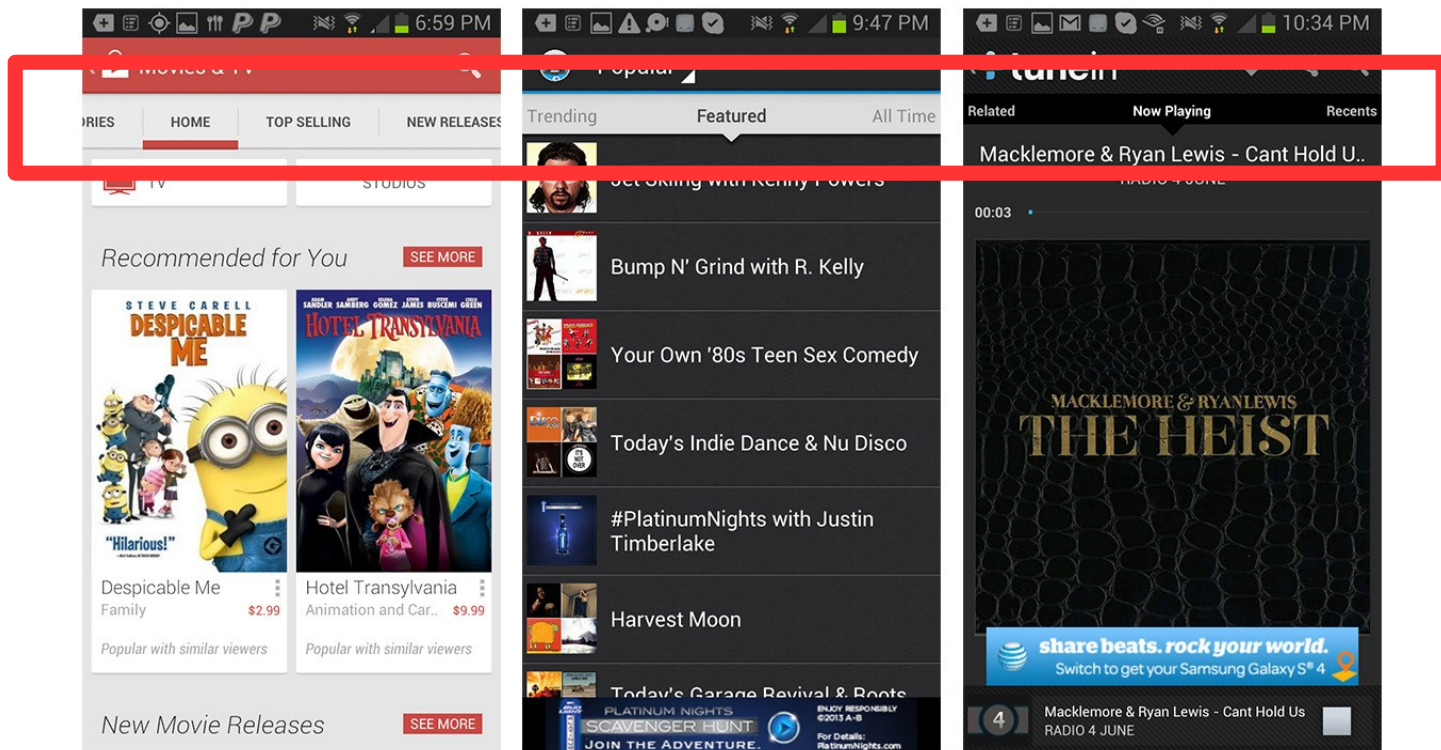


- Paging can also be based on card items

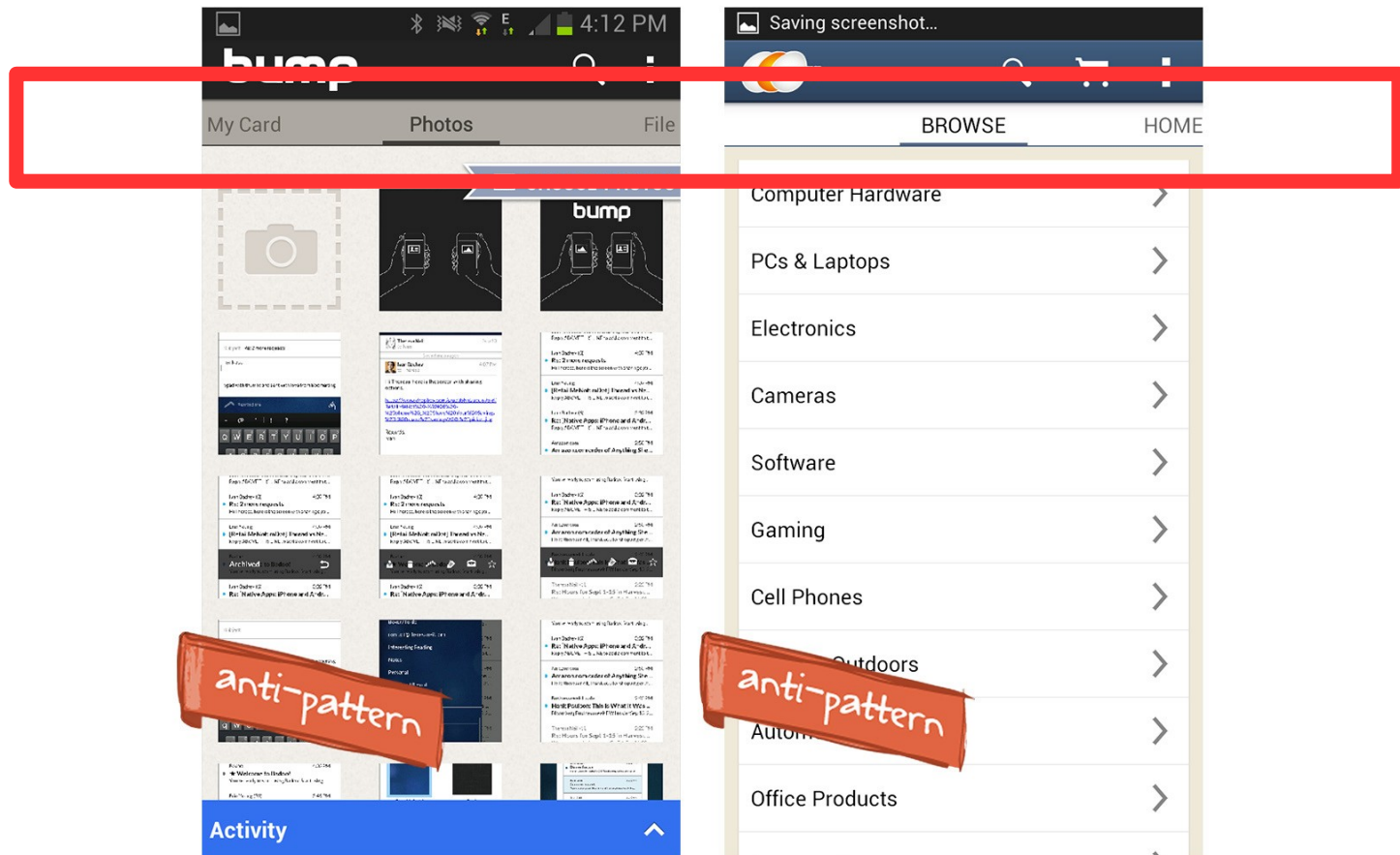


Secondary navigation: scrolling tabs

- In Android, **scrolling tabs** display a larger number of items compared to the standard fixed tabs control
- Only some of the items are displayed concurrently and swiping can be used to navigate between views

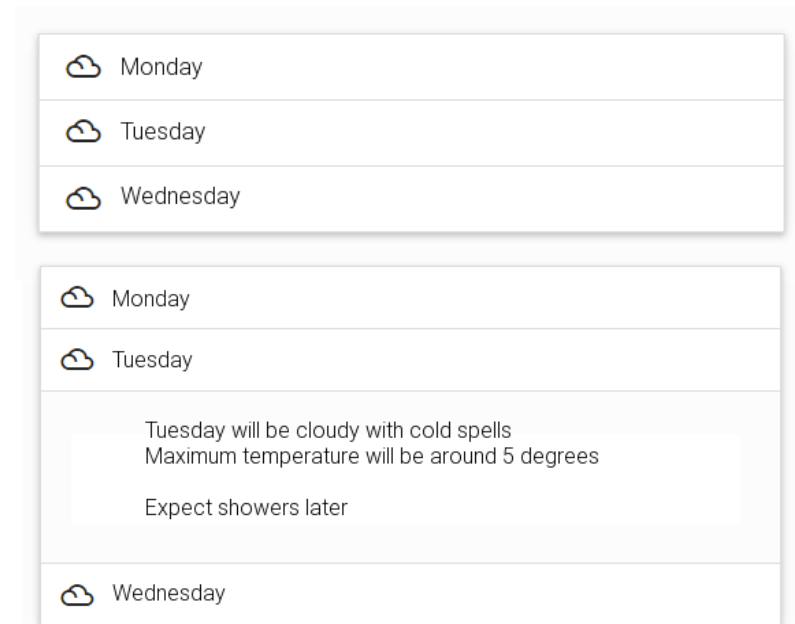
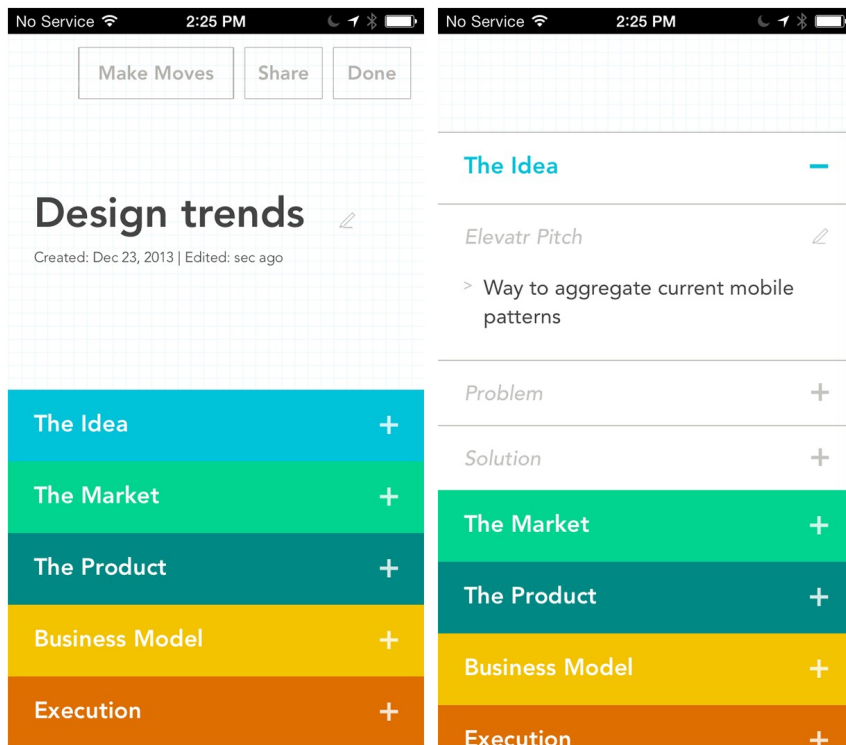


- Scrolling tabs should not be used for primary navigation



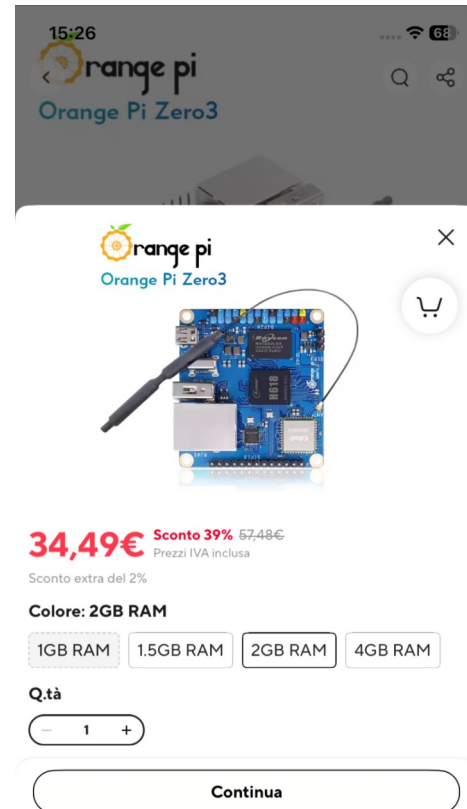
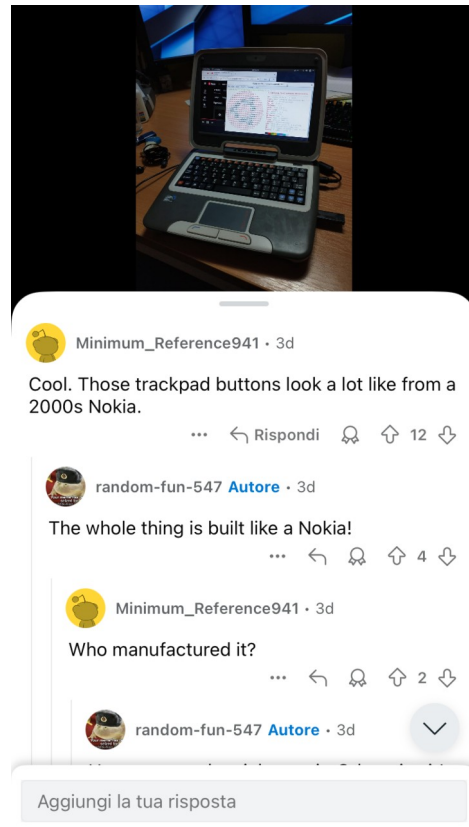
Secondary navigation: accordion

- The ***accordion*** pattern is based on displaying a stack of items that can be expanded to reveal more information while staying on the same screen



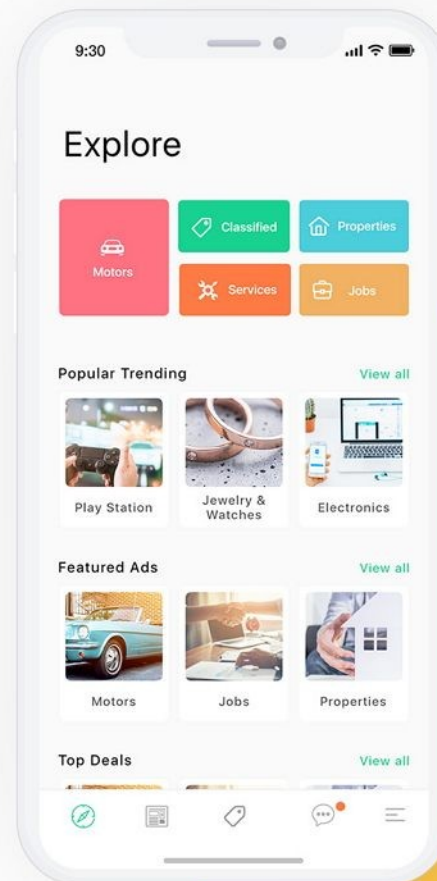
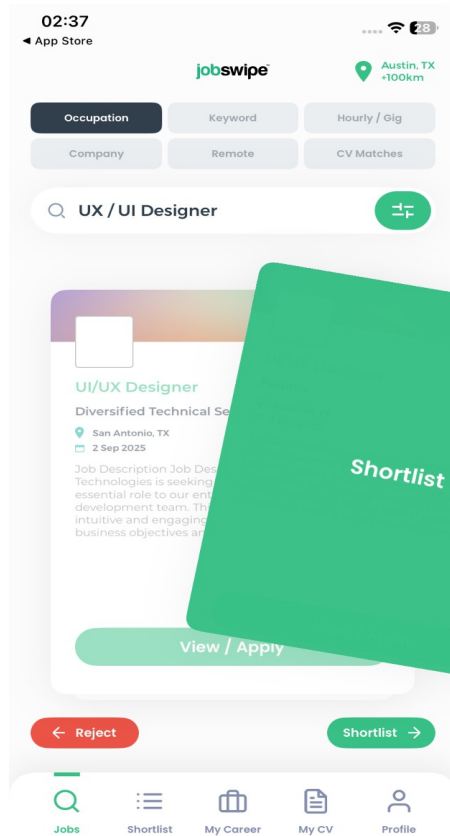
Secondary navigation: bottom sheets

- **Bottom sheets** reveal contextual content in a panel that slides up from the bottom of the screen by interacting with a visible handle



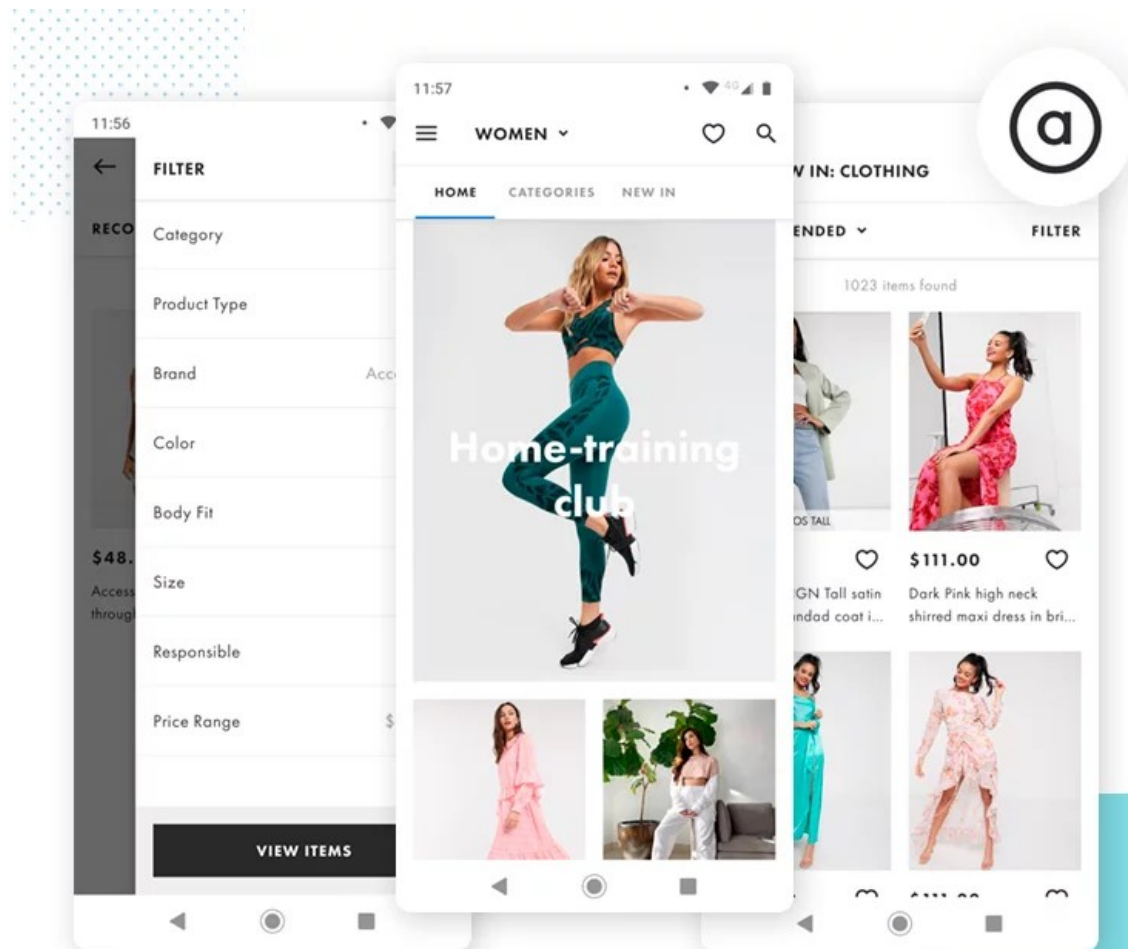
Primary + Secondary navigation

- Modern designs combine multiple primary and secondary navigation patterns



Primary + Secondary navigation

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Primary + Secondary navigation

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